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School

SIMATS ENGINEERING

SIMATS Online Programming Lab

C++ PROGRAMMING

C SANJAY 192424162A

Assignment_03_CO_01

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QUESTIONS

- Q1 Print an X-shaped number pattern 5 marks
- Q2 Check whether a number is an automorphic number 5 marks
- Q3 Find the smallest number whose digit product equals a given value 5 marks
- Q4 Print Pascal's triangle 5 marks
- Q5 Find the next number with the same digit sum 5 marks

Q1 Print an X-shaped number pattern

5 marks

Print an X pattern of size 5, with numbers on both diagonals.

1 1
2 2
3
4 4
5 5

Your Code

```
#include <iostream>

int main() {
    int n = 5;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) {

            if (i == j || i + j == n + 1) {
                std::cout << i << " ";
            } else {
                std::cout << "  ";
```

Input

stdin input

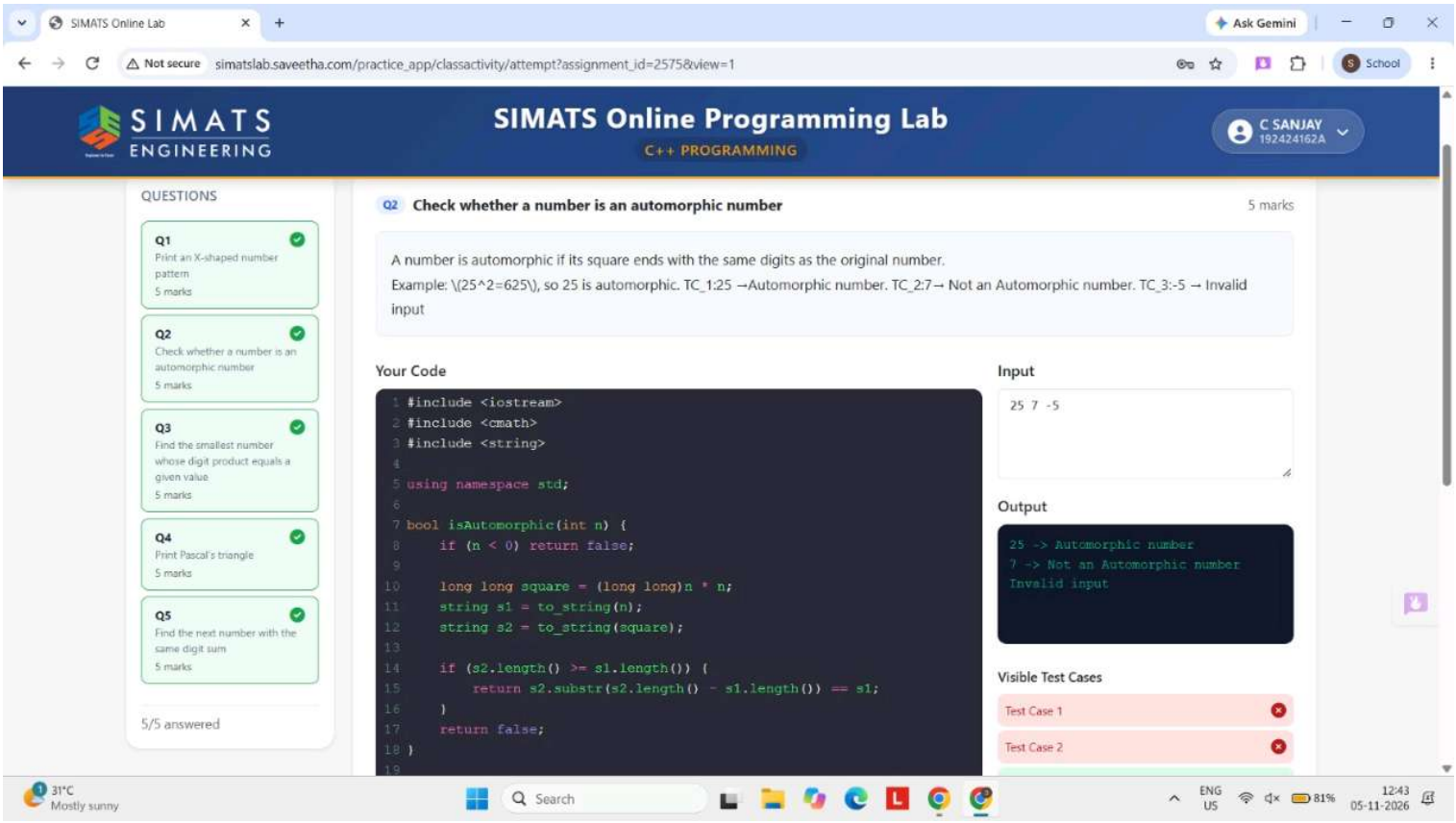
Output

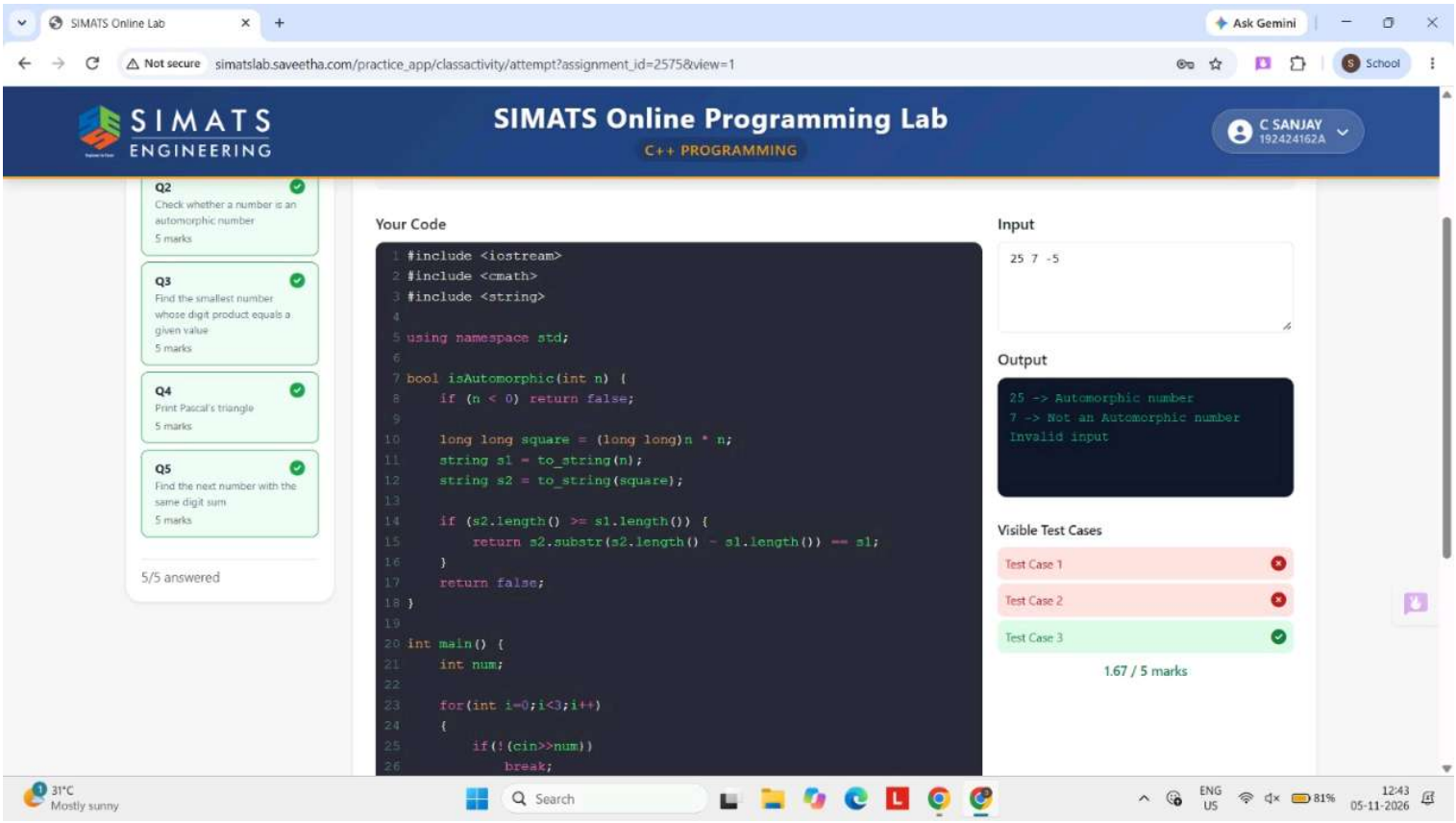
```
1       1
  2   2
    3
  4   4
```

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QUESTIONS
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Print an X-shaped number pattern
5 marks
Q2
Check whether a number is an automorphic number
5 marks
Q3
Find the smallest number whose digit product equals a given value
5 marks
Q4
Print Pascal's triangle
5 marks
Q5
Find the next number with the same digit sum
5 marks
5/5 answered
Q2 Check whether a number is an automorphic number
5 marks
A number is automorphic if its square ends with the same digits as the original number.
Example: \(25^2=625\), so 25 is automorphic. TC_1:25 →Automorphic number. TC_2:7→ Not an Automorphic number. TC_3:-5 → Invalid input
Your Code
#include <iostream>
#include <cmath>
#include <string>
using namespace std;
bool isAutomorphic(int n) {
if (n < 0) return false;
long long square = (long long)n * n;
string s1 = to_string(n);
string s2 = to_string(square);
if (s2.length() >= s1.length()) {
return s2.substr(s2.length() - s1.length()) == s1;
}
return false;
}
Input
25 7 -5
Output
25 -> Automorphic number
7 -> Not an Automorphic number
Invalid input
Visible Test Cases
Test Case 1
Test Case 2
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Q2
Check whether a number is an automorphic number
5 marks
Q3
Find the smallest number whose digit product equals a given value
5 marks
Q4
Print Pascal's triangle
5 marks
Q5
Find the next number with the same digit sum
5 marks
5/5 answered
Your Code
#include <iostream>
#include <cmath>
#include <string>
using namespace std;
bool isAutomorphic(int n) {
if (n < 0) return false;
long long square = (long long)n * n;
string s1 = to_string(n);
string s2 = to_string(square);
if (s2.length() >= s1.length()) {
return s2.substr(s2.length() - s1.length()) == s1;
}
return false;
}
int main() {
int num;
for(int i=0;i<3;i++)
{
if(!(cin>>num))
break;
Input
25 7 -5
Output
25 -> Automorphic number
7 -> Not an Automorphic number
Invalid input
Visible Test Cases
Test Case 1
Test Case 2
Test Case 3
1.67 / 5 marks
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5 marks

Q2
Check whether a number is an automorphic number
5 marks

Q3
Find the smallest number whose digit product equals a given value
5 marks

Q4
Print Pascal's triangle
5 marks

Q5
Find the next number with the same digit sum
5 marks

5/5 answered

TC_1:36 → Smallest number:49. TC_2:13 → No Solution. TC_3:0 → Smallest number:10

Your Code

```cpp
#include <iostream>
#include <vector>
#include <algorithm>
#include <string>

using namespace std;

void solve() {
    int n;
    if (!(cin >> n)) return;

    if (n == 0) {
        cout << 10 << endl;
        return;
    }

    if (n == 1) {
        cout << 1 << endl;
        return;
    }

    vector<int> digits;
    for (int i = 9; i >= 2; i--) {
        while (n % i == 0) {
```

Input

```
13
```

Output

```
No Solution
```

Visible Test Cases

Test Case 1 ✗

Test Case 2 ✓

Test Case 3 ✗

1.67 / 5 marks

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Check whether a number is an automorphic number
5 marks

Q3 ✓
Find the smallest number whose digit product equals a given value
5 marks

Q4 ✓
Print Pascal's triangle
5 marks

Q5 ✓
Find the next number with the same digit sum
5 marks

5/5 answered

1 3 3 1
1 4 6 4 1

Your Code

```cpp
#include <iostream>
using namespace std;

int main() {
    int rows = 5;
    for (int i = 0; i < rows; i++) {
        int val = 1;
        for (int j = 0; j <= i; j++) {
            cout << val << " ";
            val = val * (i - j) / (j + 1);
        }
        cout << endl;
    }
    return 0;
}
```

Input

stdin input

Output

```
1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
```

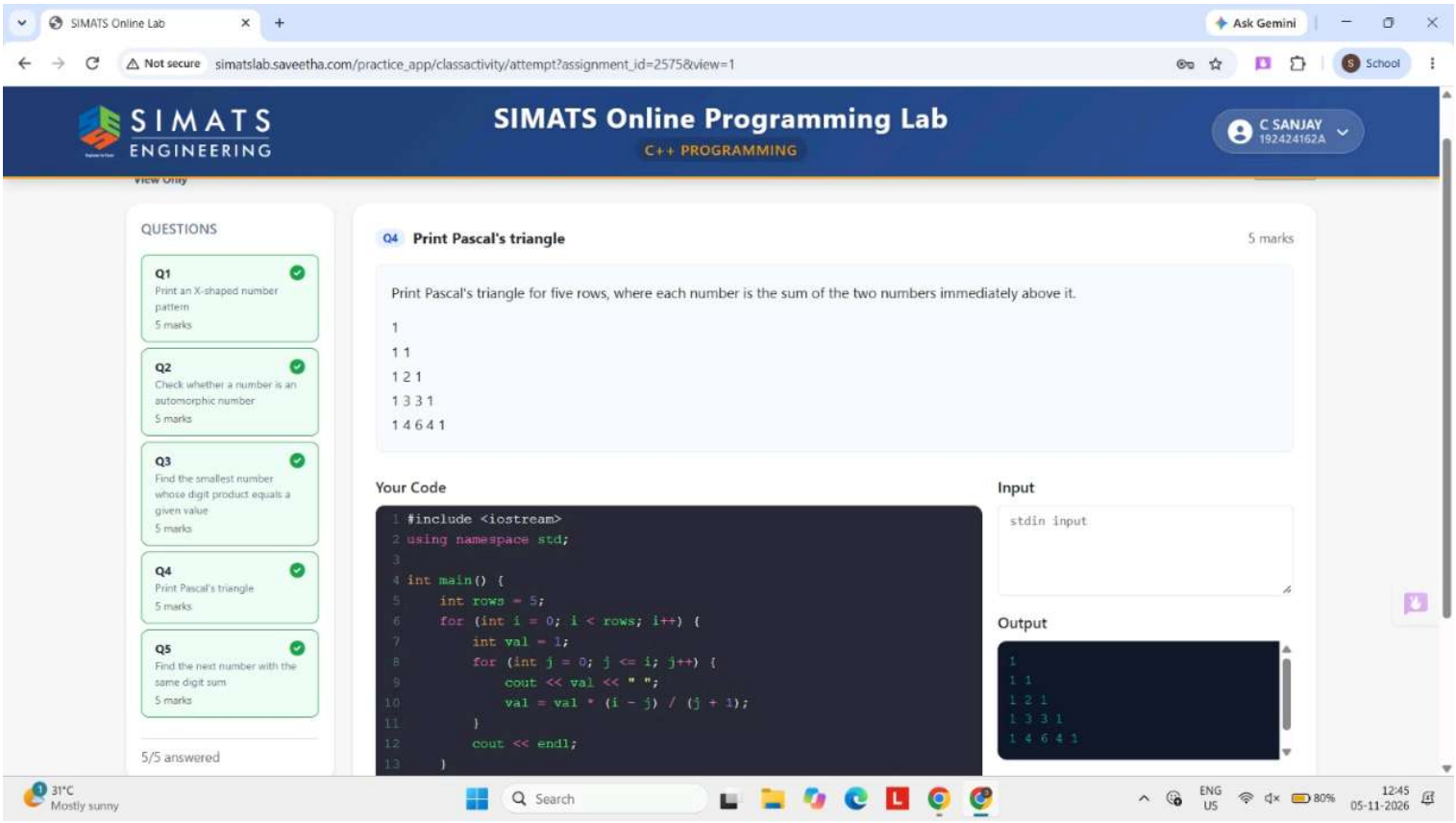
Submitted

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Q3
Find the smallest number whose digit product equals a given value
5 marks
Q4
Print Pascal's triangle
5 marks
Q5
Find the next number with the same digit sum
5 marks
5/5 answered
Q4 Print Pascal's triangle
5 marks
Print Pascal's triangle for five rows, where each number is the sum of the two numbers immediately above it.
1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
Your Code
#include <iostream>
using namespace std;
int main() {
int rows = 5;
for (int i = 0; i < rows; i++) {
int val = 1;
for (int j = 0; j <= i; j++) {
cout << val << " ";
val = val * (i - j) / (j + 1);
}
cout << endl;
}
Input
stdin input
Output
1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
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QUESTIONS

- **Q1** Print an X-shaped number pattern — 5 marks
- **Q2** Check whether a number is an automorphic number — 5 marks
- **Q3** Find the smallest number whose digit product equals a given value — 5 marks
- **Q4** Print Pascal's triangle — 5 marks
- **Q5** Find the next number with the same digit sum — 5 marks

5/5 answered

Q5 Find the next number with the same digit sum

5 marks

Read an integer and find the next greater integer having the same sum of digits.
Example: Input 59 → Output 68, because both have digit sum 14. TC_1:59 → Next number:68. TC_2:123 → Next number:132. TC_3:-0→Invalid input

Your Code

```cpp
#include <iostream>
#include <cstdlib>
using namespace std;

int getDigitSum(int n) {
    int sum = 0;
    n = abs(n);
    while (n > 0) {
        sum += n % 10;
        n /= 10;
    }
    return sum;
}

int main() {
    int num;
    int testCases = 3;
```

Input

```
59 123 0
```

Output

```
68
132
Invalid input
```

Visible Test Cases

- Test Case 1
- Test Case 2

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